

Subjective Answers for Telecom Customer Churn Prediction

1. What is the business problem you are trying to solve?

- The business objective is to predict telecom customers who are likely to churn and identify the key reasons why they churn. This will help the telecom company retain high-value customers through proactive measures.

2. What is the objective of this machine learning project?

- Build two models:

a) A model to predict whether a high-value customer will churn.

b) A model to identify important variables that are strong indicators of churn.

3. What are the primary evaluation metrics used?

- Overall accuracy is used primarily. Additionally, precision, recall, and ROC-AUC scores are analyzed to evaluate models especially under the scenario of identifying customers who will definitely churn.

4. Why use PCA?

- PCA helps reduce dimensionality while retaining maximum variance. It is suitable for the prediction model, especially with large feature space.

5. Why build a second model with feature importance?

- PCA transforms original features into non-interpretable components. Hence, a Random Forest model is built to identify key features for churn such as usage patterns, recharge behavior, etc.

6. What are the key features that cause churn?

- Based on Random Forest's feature importances, the top features causing churn include:
 - Total incoming/outgoing minutes of usage (MOU)
 - ARPU (Average Revenue per User)
 - Night calls, roaming minutes, recharge frequency, etc.

7. Business Recommendations to prevent churn:

- Offer personalized plans to high-usage customers
- Target customers with declining usage via special discounts
- Reduce call drop issues and improve network quality in churn-prone regions